

MTH401 QUIZ(3)

Lecture 33 to 41

RIZ MUGHAL
SQA ENGINEER:

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MTH401:Quiz No. 3 Quiz Start Tim

Question # 1 of 10 (Start time: 02:27:03 PM, 18 February 2021)

This equation

$$\det(A - \lambda I)$$

is known as _____of the matrix A.

Select the correct option

○	Linear equation
○	Algebraic equation
●	Characteristic equation
○	All of these

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MTH401:Quiz No. 3 Quiz Start Time: 02:27 PM, 18 Febru

Question # 2 of 10 (Start time: 02:27:21 PM, 18 February 2021) Total

The eigenvalues of matrix

$$A = \begin{pmatrix} 3 & 4 \\ -1 & 7 \end{pmatrix}$$

Select the correct option

Reload Math Equ

- [x] $\lambda_1 = 5, \lambda_2 = 5$
- [ ] $\lambda_1 = 3, \lambda_2 = 7$
- [ ] $\lambda_1 = 5, \lambda_2 = -5$
- [ ] $\lambda_1 = -5, \lambda_2 = -5$

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MTH401:Quiz No. 3 Quiz Start Time: 02:27 PM, 18 February 2021

Question # 3 of 10 (Start time: 02:27:37 PM, 18 February 2021) Total Marks: 1

The Legendre polynomials can also be generated by Rodrigues formula, $P_n(x) = \frac{1}{2^n n!} \frac{d^n}{dx^n}(x^2 - 1)^n$.

Select the correct option

Reload Math Equations

- (●) True
- ( ) False

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Click to Save Answer & Move to Next Question

MTH401:Quiz No. 3 Quiz Start Time: 02:27 PM, 18 February 2021

Question # 4 of 10 (Start time: 02:27:51 PM, 18 February 2021) **Total Marks: 1**

Operator method or systematic elimination method of solution of a system of linear homogeneous or linear non-homogeneous differential equations provides us _____ differential equation in one of the dependent variables that has not been eliminated.

Select the correct option

- ○ many
- ◉ single

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MTH401:Quiz No. 3 Quiz Start Time: 02:27 PM, 18 February 2021

Question # 5 of 10 (Start time: 02:28:06 PM, 18 February 2021) Total Marks: 1

Legendre polynomials are normal with respect to the weight function

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Select the correct option

Reload Math Equations

- [x] $w(x) = \frac{2n+1}{n}$
- [ ] $w(x) = \frac{n+1}{2n}$
- [ ] $w(x) = \frac{2n^2+1}{n}$
- [ ] $w(x) = \frac{2n+1}{n^2}$

MTH401:Quiz No. 3 Quiz Start Time: 02:27 PM, 18 February

Question # 6 of 10 (Start time: 02:28:21 PM, 18 February 2021) Total Ma

The equation, $\det(A - \lambda I) = 0$, is known as the characteristic equation.

Select the correct option

Reload Math Equati

○	False
●	True

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MTH401:Quiz No. 3 Quiz Start Time: 02:27 PM, 18 F

Question # 7 of 10 (Start time: 02:28:34 PM, 18 February 2021)

It is NOT always possible to solve the given system for the highest-order derivative of each dependent variable.

Select the correct option

- ○ False
- ◉ True

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MTH401:Quiz No. 3 Quiz Start Time: (

Question # 8 of 10 (Start time: 02:28:49 PM, 18 February 2021)

The horizontal lines in a matrix are called the _____ of the matrix.

Select the correct option

- columns
- rows

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Question # 9 of 10 (Start time: 02:29:06 PM, 18 February 2021)

x=0 is a(an) -------- point of the differential equation: $y'' + xy' + 2y = 0.$

Select the correct option

○	Singular
○	Regular singular
○	Irregular singular
◉	Ordinary

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MTH401:Quiz No. 3 Quiz Start Time: 02:27 PM, 18 February 2021

Question # 10 of 10 (Start time: 02:29:23 PM, 18 February 2021) Total Marks: 1

If

$$X_1 = \begin{pmatrix} e^{-2t} \\ -e^{-2t} \end{pmatrix}$$

Select the correct option

Reload Math Equations

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- $\begin{pmatrix} -2e^{-2t} \\ -2e^{-2t} \end{pmatrix}$
- $\begin{pmatrix} 2e^{-2t} \\ -2e^{-2t} \end{pmatrix}$
- $\begin{pmatrix} 2e^{-2t} \\ 2e^{-2t} \end{pmatrix}$
- (selected) $\begin{pmatrix} -2e^{-2t} \\ 2e^{-2t} \end{pmatrix}$

2nd account

MTH401:Quiz No. 3 Quiz Start Time: 02:31 PM

Question # 1 of 10 (Start time: 02:31:24 PM, 18 February 2021)

Vectors $\begin{pmatrix} -2 \\ 3 \end{pmatrix}$ and $\begin{pmatrix} -66 \\ 99 \end{pmatrix}$ are linearly $- - - - -$.

Select the correct option

Reload

- (●) Dependent
- ( ) Independent

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MTH401:Quiz No. 3 Quiz Start Time: 02:31 PM, 18 February 2021

Question # 2 of 10 (Start time: 02:31:40 PM, 18 February 2021) **Total Marks:**

Operator method or systematic elimination method of solution of a system of linear homogeneous or linear non-homogeneous differential equations is based on the process of systematic elimination of the _____ variables.

Select the correct option

- ○ independent
- ◉ dependent

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MTH401:Quiz No. 3 Quiz Start

Question # 3 of 10 (Start time: 02:31:54 PM, 18 February 2021)

Spherical Bessel functions are used in many physical problems.

Select the correct option

- ○ False
- ◉ True

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MTH401:Quiz No. 3 Quiz Start Time: 02:31 PM, 18 I

Question # 4 of 10 (Start time: 02:32:07 PM, 18 February 2021)

If

$$X_1 = \begin{pmatrix} e^{-2t} \\ -e^{-2t} \end{pmatrix}$$

Select the correct option

Reload Mat

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○	$\begin{pmatrix} -2e^{-2t} \\ -2e^{-2t} \end{pmatrix}$
○	$\begin{pmatrix} 2e^{-2t} \\ -2e^{-2t} \end{pmatrix}$
○	$\begin{pmatrix} 2e^{-2t} \\ 2e^{-2t} \end{pmatrix}$
●	$\begin{pmatrix} -2e^{-2t} \\ 2e^{-2t} \end{pmatrix}$

MTH401:Quiz No. 3 Quiz Start Time: 02:31 PM, 18 February 20

Question # 5 of 10 (Start time: 02:32:21 PM, 18 February 2021) Total Marks

If n is positive integer, then the solution of Legendre's differential equation is called a Legendre's _____ of degree n.

Select the correct option

Reload Math Equation

- function
- polynomial

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MTH401:Quiz No. 3 Quiz Sta

Question # 6 of 10 (Start time: 02:32:35 PM, 18 February 2021)

Solution of Legendre's differential equation is called

Select the correct option

- (●) Legendre function.
- ( ) Legendre polynomial.
- ( ) Legendre differential.
- ( ) Recurrence relation.

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MTH401:Quiz No. 3 Quiz Start Time: 02:31 PM, 18 February 202

Question # 7 of 10 (Start time: 02:32:49 PM, 18 February 2021) Total Marks:

The matrix form of non - homogeneous system of differential equationis

$$X' = -3x + 4y - 9z$$
$$X' = 6x - y$$
$$X' = 10x - 4y + 3z$$

Select the correct option

Reload Math Equations

- $X' = \begin{bmatrix} -3 & -9 & -4 \\ 6 & 0 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$
- $X' = \begin{bmatrix} -3 & 4 & -9 \\ 6 & -1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$
- $X' = \begin{bmatrix} -3 & 4 & x \\ 6 & -1 & y \end{bmatrix} \begin{bmatrix} -9 \\ 0 \end{bmatrix}$
- $X' = \begin{bmatrix} -3 & 4 & -9 \\ \end{bmatrix} \begin{bmatrix} x \\ \end{bmatrix}$

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MTH401:Quiz No. 3 Quiz Start Time: 02:31 PM, 18 February 202

Question # 8 of 10 (Start time: 02:33:12 PM, 18 February 2021) Total Marks:

Every n^{th} order differential equation, $y^{(n)} = F\left(t, y, y', \ldots, y^{(n-1)}\right)$, as well as most systems of differential equations, could be reduced to the _____ order system.

Select the correct option

Reload Math Equations

- $n-1$
- $n+1$
- n^{th}
- infinite

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MTH401:Quiz No. 3 Quiz Start Time: 02:31 PM, 18 February 2021

Question # 9 of 10 (Start time: 02:33:26 PM, 18 February 2021) **Total Marks: 1**

The simultaneous ordinary differential equations involve two or more equations that contain derivatives of two or more unknown functions of _____ independent variable.

Select the correct option

Reload Math Equations

- [x] 1
- [ ] 2
- [ ] 2
- [ ] infinite

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MTH401:Quiz No. 3 Quiz Start Time: 02:31 PM, 18 February 2021

Question # 10 of 10 (Start time: 02:33:40 PM, 18 February 2021) Total Marks: 1

The Legendre polynomials can also be generated by Rodrigues formula, $P_n(x) = \frac{1}{2^n\, n!}\frac{d^n}{dx^n}(x^2-1)^n$.

Select the correct option

Reload Math Equations

- (●) True
- ( ) False

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Click to Save Answer & Move to Next Question

3rd account

MTH401:Quiz No. 3 Quiz Start Time: 03:09 F

Question # 1 of 10 (Start time: 03:09:28 PM, 18 February 2021)

The solution of Bessel equation is denoted by _____.

Select the correct option

Relo

- $B_v(x)$
- $J_v(x)$
- $P_v(x)$
- $L_v(x)$

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MTH401:Quiz No. 3 Quiz Start Time: 03:09 PM, 18 Fet

Question # 2 of 10 (Start time: 03:09:44 PM, 18 February 2021) To

The equation form of non - homogeneous system of differential equationis

$$X' = \begin{bmatrix} -2 & 4 \\ & \end{bmatrix} \begin{bmatrix} x \\ \end{bmatrix} + \begin{bmatrix} 1 \\ \end{bmatrix} e^t + \begin{bmatrix} -4 \\ \end{bmatrix} t$$

Select the correct option

Reload Math E

◉	$X' = -2x + 4y + e^t - 4t$ $X' = 3x - 4y + 11t$
○	$X' = -2x - 4y + e^t - 4t$ $X' = 3x + 4y + 11t$
○	$X' = 2x + 4y + e^t - 4t$ $X' = 3x - 4y + 11t$
○	$X' = 2x + 4y - e^t - 4t$ $X' = 3x - 4y - 11t$

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MTH401:Quiz No. 3

Quiz Start Time: 03:09 PM, 18 February 2021

Question # 3 of 10 (Start time: 03:09:57 PM, 18 February 2021)

Total Marks: 1

The non - trivial solution of the system of differential equation exist only when

Select the correct option

Reload Math Equations

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- $del(X - \lambda I) = 0$
- $del(A - \lambda I) \neq 0$
- $del(A + \lambda I) = 0$
- $del(A - \lambda I) = 0$

MTH401:Quiz No. 3 Quiz Start Time: 03:09 PM, 18 February 20

Question # 4 of 10 (Start time: 03:10:10 PM, 18 February 2021) Total Marks

A solution of a system of differential equations is a set of differentiable functions, $x = f(t),\ y = g(t),\ x = h(t), \ldots,$ those satisfy _____ equation(s) of the system on some interval I.

Select the correct option

Reload Math Equation

- one
- at least one
- two
- each

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MTH401:Quiz No. 3 Quiz Start Time: 03:09 PM, 18 February 2021

Question # 5 of 10 (Start time: 03:10:26 PM, 18 February 2021) **Total Marks:** 1

An immediate consequence of the principle of superposition is that a constant multiple of any solution vector of a homogeneous system of first order differential equation is also a solution of the system.

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Select the correct option

- (●) True
- ( ) False

MTH401:Quiz No. 3 Quiz

Question # 6 of 10 (Start time: 03:10:42 PM, 18 February 2021)

Recurrence relation for polynomials relates

Select the correct option

- (●) Polynomials of different degrees.
- ( ) Polynomials of same degrees.
- ( ) Equal polynomials.
- ( ) All options are true.

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MTH401:Quiz No. 3 Quiz Start Time:

Question # 7 of 10 (Start time: 03:10:54 PM, 18 February 2021)

The solution of Legendre differential equation is denoted by _____.

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Select the correct option

○	$B_v(x)$
○	$J_v(x)$
●	$P_v(x)$
○	$L_v(x)$

MTH401:Quiz No. 3 Quiz Start Time: 03:09 PM.

Question # 8 of 10 (Start time: 03:11:09 PM, 18 February 2021)

Legendre polynomials are normal with respect to the weight function

Select the correct option

Reload

- [x] $w(x) = \frac{2n+1}{n}$
- [ ] $w(x) = \frac{n+1}{2n}$
- [ ] $w(x) = \frac{2n^2+1}{n}$
- [ ] $w(x) = \frac{2n+1}{n^2}$

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MTH401:Quiz No. 3 Quiz Start Time: 03:09 PM, 18

Question # 9 of 10 (Start time: 03:11:22 PM, 18 February 2021)

Legendres polynomials are specific _____ degree polynomials.

Select the correct option

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○	$n-1$
○	$n+1$
●	n
○	n^2

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MTH401:Quiz No. 3 Quiz Start Time: 03:09 PM, 18

Question # 10 of 10 (Start time: 03:11:36 PM, 18 February 2021)

Bessel equation occurs frequently in advances studies in _____.

Select the correct option

- applied mathematics
- physics
- engineering
- All of these

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Thank you for watching ☺

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rizwanqadeer848@gmail.com